

AN ECLECTIC TECHNOLOGIES REPORT

BES Asset Nomenclature Specification

A Comprehensive Framework for Bulk Electric System Asset Identification

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Tim Roxey

Eclectic Technologies

tim.roxey@gmail.com · 410.474.9240 (mobile/Signal)

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Executive Summary

This specification establishes a comprehensive nomenclature framework for identifying assets within the Bulk Electric System as defined by NERC standards. The framework addresses the fundamental challenge of creating asset identifiers that serve both human operators and automated compliance systems while maintaining consistency across diverse utility ownership structures, operational contexts, and legacy naming conventions inherited from decades of industry evolution.

The nomenclature system defined herein supports multiple simultaneous objectives that have historically been addressed through incompatible approaches. It enables human operators to quickly identify asset type, location, and operational function through recognizable patterns while simultaneously encoding machine-parsable metadata that supports automated correlation across cybersecurity threat intelligence, NERC compliance evidence collection, and operational data integration. The framework accommodates both hierarchical asset relationships, such as the containment of protection relays within substations, and lateral associations, such as the termination points of transmission lines across organizational boundaries.

Implementation of this nomenclature standard directly supports the ERT Companion system's automated evidence collection capabilities by establishing consistent asset references that can be correlated across operational technology environments, information technology security controls, and regulatory compliance documentation. The framework further enables integration with the Analyst's WorkBench threat intelligence correlation engine via standardized asset-classification metadata, supporting automated identification of affected infrastructure when vulnerabilities or threats are reported in vendor advisories or intelligence reports.

The specification recognizes that nomenclature standardization represents a significant operational investment for entities with established legacy naming systems. Accordingly, the framework defines both strict compliance requirements for newly commissioned assets and pragmatic transition approaches for existing infrastructure. The specification prioritizes elements essential for automated compliance and security functions while providing flexibility in components that serve primarily operational purposes within individual entities.

1. Introduction and Purpose

Asset identification within the Bulk Electric System has evolved organically across decades of utility operations, reflecting the technical constraints, organizational practices, and operational priorities of different eras. Early-generation facilities often received names reflecting their fuel sources or geographic locations, while transmission assets were identified by circuit designations tied to protection relay schemes developed when electromechanical relays were the primary coordination mechanism. Substations inherited naming conventions from the communities they served or the transmission corridors they interconnected. This historical evolution has produced a rich but inconsistent landscape of identification approaches across the industry.

The emergence of NERC Critical Infrastructure Protection standards introduced new requirements for asset tracking and security control implementation that depend upon clear identification of BES assets and their attributes. These compliance obligations intersect with cybersecurity needs arising from the integration of information technology systems with operational technology environments. Modern threat intelligence correlation requires the ability to rapidly identify all instances of vulnerable components across an entity's infrastructure, a capability that depends on systematic asset identification that encodes sufficient metadata to support automated queries.

This specification responds to three distinct but interrelated needs. First, it addresses the operational requirement for asset identifiers that enable rapid comprehension for system operators, protection engineers, and field personnel responding to system events under time pressure. Second, it satisfies the compliance requirement for consistent asset references across documentation, from network architecture diagrams to evidence submissions for NERC audits. Third, it enables automated correlation capabilities required in modern cybersecurity operations, where threat intelligence must be rapidly translated into infrastructure-specific impact assessments.

The framework presented in this document establishes nomenclature rules: mandatory requirements for core identification elements, recommended practices for commonly encountered asset attributes, and optional conventions for entity-specific extensions. This tiered approach recognizes that an excessive prescription would either fail to accommodate legitimate operational diversity or be ignored in practice, while insufficient structure would fail to achieve the interoperability objectives that motivate standardization efforts.

2. Design Principles and Requirements

The nomenclature framework is based on several foundational design principles that inform specific implementation choices throughout the specification. These principles reflect hard-won lessons from previous standardization attempts and the practical realities of operating large-scale electric infrastructure.

2.1 Human Comprehension Priority

Asset identifiers must remain comprehensible to human operators who encounter them during normal operations and emergency response situations. This requirement takes precedence over theoretical optimizations for machine parsing. An identifier that requires database lookups or documentation references to interpret is insufficiently transparent for operational use. The framework therefore employs recognizable abbreviations drawn from established industry practice rather than arbitrary numeric codes, even when numeric schemes might offer slightly superior database efficiency. System operators must be able to understand that RIVR-GEN1-GT3 refers to the third gas turbine at the River Bend generation facility without consulting reference materials.

2.2 Hierarchical Relationship Encoding

BES assets exist within hierarchical containment relationships that must be evident from their identifiers. A protection relay is located within a substation bay, which is part of a substation facility. A generating unit is part of a generation plant, which may be one of several plants operated by a registered entity. The nomenclature structure employs positional encoding, where earlier identifier components represent higher-level containers and later components represent increasingly specific assets. This enables automated systems to extract containment relationships via simple string parsing. The identifier WSTFLD-SUB.B3.REL-51N immediately conveys that relay 51N is within bay B3 of the Westfield substation without requiring relationship tables.

2.3 Unique Instance Identification

Every physical asset that represents a distinct failure point or security boundary must receive a unique identifier. This requirement extends to assets that are identical in specification but deployed separately. Four protection relays of identical model and configuration, protecting different circuit breakers, each receive a distinct identifier because they represent four separate points at which failure or compromise could impact system operations. Conversely, redundant power supplies within a single protection relay do not receive separate identifiers because they do not represent separate operational boundaries. The distinction rests on whether independent failure modes exist that would require distinct incident response or maintenance activities.

2.4 Classification Metadata Integration

Asset identifiers must encode sufficient classification metadata to support automated queries without requiring lookup operations against external databases. This requirement arises from the operational contexts where identifier interpretation occurs. During cybersecurity incident response, analysts must rapidly identify all assets potentially affected by a newly disclosed vulnerability. This requires the ability to filter assets by technology type, vendor, and operational function. The nomenclature framework

embeds this information through systematic component selection and positional encoding. The identifier METRO-CC.EMS-SRV indicates that this is a server within the energy management system at the Metro control center, enabling automated identification of EMS infrastructure to support vendor patch applicability assessment.

2.5 Delimiter Consistency and Parsing Rules

The framework employs a consistent delimiter hierarchy that supports automated parsing while maintaining human readability. Hyphens separate major identifier components representing different hierarchy levels or classification dimensions. Periods separate sub-components within a hierarchy level. Underscores connect multi-word descriptors that function as single logical units. This structured approach to delimiter usage enables simple parsing rules, allowing automated systems to split identifiers on hyphens to extract major components, then split those components on periods to access finer granularity. The consistency of this delimiter usage across all asset types eliminates the need for asset-type-specific parsing logic.

2.6 Namespace Collision Avoidance

Within a registered entity's scope, asset identifiers must be globally unique and require no coordination across organizational units. This requirement prevents deployment delays caused by identifier approval processes and reduces the operational burden of nomenclature management. The framework achieves namespace isolation by requiring the inclusion of facility identifiers in all asset names. Assets across different facilities are assigned different identifiers, regardless of other attributes. This approach trades slightly longer identifiers for dramatically simplified identifier assignment and the elimination of collision risks that would otherwise require resolution protocols.

3. Nomenclature Architecture Overview

BES asset identifiers follow a consistent structural pattern regardless of asset type. This consistency enables both human operators and automated systems to extract meaning from identifiers via pattern recognition rather than through exhaustive, type-specific rules. The architecture defines five primary identifier components arranged in hierarchical order from broadest to most specific scope.

3.1 Core Identifier Structure

Every BES asset identifier conforms to the following pattern: ENTITY-FACILITY-SYSTEM.SUBSYSTEM.COMPONENT. The ENTITY component identifies the registered entity responsible for the asset under NERC standards, enabling automated correlation of assets to compliance obligations and organizational boundaries. The FACILITY component identifies the physical location or operational grouping to which the asset belongs. The SYSTEM component identifies the primary functional system classification. The SUBSYSTEM component provides additional functional specificity when multiple subsystems exist within a system category. The COMPONENT identifier uniquely identifies the specific asset instance within its subsystem context.

Not all components are required for every asset type. Generation assets typically employ a three-component pattern of FACILITY-PLANT-UNIT since they do not require subsystem granularity. Transmission lines employ facility-to-facility naming that captures both terminal points. Control system assets may extend the subsystem hierarchy to accommodate complex network topologies. The framework defines mandatory and optional components for each asset category in subsequent sections.

3.2 Entity Identifier Conventions

Entity identifiers distinguish between assets owned by different registered entities while maintaining brevity for common operational usage. For entities with established industry-wide abbreviations, those abbreviations may be used directly. Duke Energy becomes DUKE, Southern Company becomes SO, and Pacific Gas and Electric becomes PGE. For entities without recognized abbreviations, a four-to-six-character mnemonic should be derived from the entity name using consonant extraction and vowel reduction techniques. Northern States Power Company might become NRTHT or NSP depending on regional recognition patterns.

Entities operating across multiple regulatory regions may incorporate regional designators where beneficial for operational clarity. An entity operating in both WECC and ERCOT might use entity identifiers such as ENTCO-W and ENTCO-T to distinguish assets registered in different interconnections when this distinction matters for operational coordination. This practice remains optional and should be employed only where regional distinctions carry operational significance.

3.3 Facility Identifier Conventions

Facility identifiers represent the primary location-based component of asset names. For generation facilities, this typically reflects the plant name derived from geographic, fuel source, or historical naming conventions. For substations, facility identifiers should employ abbreviations of the substation name, typically four to eight characters that preserve recognition while maintaining brevity. Transmission

facilities employ a different pattern capturing both terminal substations. Control centers receive facility identifiers reflecting their operational scope or geographic service area.

Facility identifier selection should prioritize stability over precision. A facility identifier should remain constant even if the legal name of the location changes due to administrative reorganization or geographic boundary adjustments. The facility identifier serves as a stable reference point for historical trending, maintenance records, and compliance documentation. Changing facility identifiers disrupts this continuity and should be avoided except where physical assets are decommissioned and replaced with fundamentally different infrastructure at a new location.

3.4 Voltage Class Integration

Transmission and substation assets incorporate voltage class as a mandatory identifier component because voltage determines fundamental operating characteristics, protection requirements, and regulatory applicability. Voltage designations use nominal system voltages: 765, 500, 345, 230, 161, 138, 115, 69. These values follow ANSI standard voltage classes and represent the nominal line-to-line voltage in kilovolts. Assets serving multiple voltage levels receive compound designators, such as 345-230, to indicate transformation or interconnection functions.

The voltage component appears immediately after the facility identifier for transmission assets, establishing the voltage context before functional components appear. This positioning reflects the fundamental role of voltage class in transmission operations, where voltage is the primary classification axis that determines applicable standards, clearance requirements, and operational procedures. For substations, voltage appears within the system designation to distinguish equipment associated with different voltage transformation stages.

4. Generation Asset Nomenclature

Generation facilities present nomenclature challenges arising from the diversity of prime mover technologies, fuel sources, operational modes, and ownership arrangements. A combined-cycle natural gas plant operates fundamentally differently from a nuclear station or a photovoltaic array, yet the nomenclature framework must accommodate all generation types within a coherent structure. The approach taken here establishes common patterns for facility and unit identification while incorporating technology-specific designators that capture operationally significant distinctions.

4.1 Generation Facility Identifiers

Generation facility identifiers should reflect established plant names while conforming to length and character set constraints. Names like Riverbend, Mountain View, Prairie Island convey geographic context that supports operational communication. Where facilities serve multiple generating technologies or configurations, the base facility identifier should represent the facility as a whole rather than any particular technology deployed there. A facility that includes both combustion turbines and combined-cycle generation receives a single facility identifier, with technology distinctions encoded in subsequent identifier components.

Multi-site generation portfolios, such as wind or solar installations distributed across geographic areas, require a systematic assignment of facility identifiers that balances geographic specificity with manageable proliferation. Large solar installations may be subdivided into facility areas, each assigned a distinct facility identifier when the subdivisions represent meaningful operational or administrative boundaries. Wind farms typically receive facility identifiers at the project level rather than at the individual turbine-cluster level, unless clusters operate under separate interconnection agreements or ownership arrangements.

4.2 Generation Technology Designators

The system component of generation asset identifiers employs standardized technology codes that indicate prime mover type and general operational characteristics. These designators support automated identification of generation technology for purposes ranging from emissions reporting to cybersecurity vulnerability assessment, where different technologies employ different control systems with different vulnerability profiles.

Technology Code	Prime Mover Type
COAL	Coal-fired steam turbine generation
NGCC	Natural gas combined-cycle generation
CT	Combustion turbine (simple-cycle peaking)
NUCLEAR	Nuclear steam turbine generation
HYDRO	Conventional hydroelectric generation
PUMPED	Pumped storage hydroelectric

WIND	Wind turbine generation
SOLAR-PV	Photovoltaic solar generation
SOLAR-T	Concentrating solar thermal generation
BATTERY	Battery energy storage systems
DIESEL	Diesel reciprocating engine generation
GEO	Geothermal generation

These technology designators appear in the system position of the identifier structure: FACILITY-TECH-UNIT. For a combined-cycle facility, this produces identifiers such as RIVRBND-NGCC-U1 for combined-cycle unit 1 and RIVRBND-CT-GT5 for combustion turbine 5 when operated in simple-cycle peaking mode. This approach accommodates facilities with multiple technology types while maintaining consistent identifier patterns.

4.3 Generating Unit Identifiers

Individual generating units receive numeric or alphanumeric designators that uniquely identify each unit within its technology and facility context. Traditional large thermal plants typically use simple sequential numbering, such as U1, U2, U3, for steam turbine units. Combined-cycle facilities may use component-specific prefixes: GT1, GT2 for gas turbines; ST1 for steam turbines; HRSG1, HRSG2 for heat recovery steam generators. This level of granularity supports maintenance management and cybersecurity assessment, where different component types may employ different control systems.

Renewable generation facilities face particular challenges in unit identification due to the large number of individual generating assets. A wind farm might include one hundred turbines; a large solar installation might include thousands of inverters. The nomenclature framework accommodates this through hierarchical grouping. Wind turbines receive identifiers that encode both their facility area and specific turbine number: WINDS-WIND-N.WT15 indicates wind turbine 15 in the north area of the Windscape facility. Solar inverters similarly employ zone-based identification: SUNNYSIDE-SOLAR_PV-E.INV045 identifies inverter 45 in the east zone of the Sunnyside solar facility.

4.4 Generation Asset Examples

The following examples illustrate complete generation asset identifiers conforming to the nomenclature framework.

Asset Identifier	Asset Type	Description
RIVRBND-COAL-U1	Coal Unit	Coal-fired unit 1 at the Riverbend facility
RIVRBND-COAL-U1.BOILER	Boiler	Boiler system for coal unit 1

RIVRBND-COAL-U1.GEN	Generator	Generator for coal unit 1
MTVIEW-NGCC-GT1	Combustion Turbine	Gas turbine 1 at Mountain View NGCC
MTVIEW-NGCC-ST1	Steam Turbine	Steam turbine at Mountain View NGCC
MTVIEW-NGCC-HRSG1	HRSG	Heat recovery steam generator 1
PRAIRIE-NUCLEAR-U1	Nuclear Unit	Nuclear Unit 1 at Prairie Island
RIVERSIDE-HYDRO-U3	Hydro Unit	Hydroelectric Unit 3 at Riverside
WINDS-WIND-N.WT032	Wind Turbine	Wind turbine 32 in the north section
SUNNY-SOLAR_PV-W.INV154	Solar Inverter	Inverter 154 in the west section
PEAK-CT-GT8	Peaking Unit	Combustion turbine 8 at Peak facility
STORAGE-BATTERY-BESS1	Battery Storage	Battery energy storage system 1

5. Transmission Asset Nomenclature

Transmission assets connect generation facilities to load centers and interconnect different portions of the BES. Unlike generation assets, which exist at discrete facilities, transmission lines span distances and connect facilities operated by different entities. The nomenclature framework must accommodate this distributed nature while supporting operational requirements for circuit identification, protection coordination, and regulatory compliance tracking.

5.1 Transmission Line Naming Conventions

Transmission line identifiers capture three essential attributes: terminal substations, voltage class, and circuit designation. The basic pattern follows SUBSTATION1-SUBSTATION2-VOLTAGE-CIRCUIT, where the substation names identify the line terminals, voltage indicates the operating voltage class, and circuit distinguishes multiple lines operating between the same terminal pairs. This pattern produces identifiers such as WSTFLD-METRO-345-L1 for the first 345kV line connecting the Westfield substation to the Metro substation.

Substation ordering in transmission line identifiers should follow consistent conventions to avoid ambiguity where multiple entities might reference the same line. The convention recommended here lists substations in alphabetical order unless operational practice strongly favors directional naming that reflects power flow patterns. For lines connecting generation facilities to transmission substations, the generation facility appears first: RIVRBND-WSTFLD-345-G1 indicates a generator lead from the Riverbend facility to the Westfield substation.

5.2 Circuit Designations and Multiple Lines

Multiple transmission lines operating between the same terminal substations at the same voltage require distinct circuit designations. Traditional utility practice employs numeric suffixes: L1, L2, L3. Some utilities employ directional designations or compass references where lines follow different physical paths: WSTFLD-METRO-345-NORTH and WSTFLD-METRO-345-SOUTH. The framework supports both approaches but recommends simple numeric designations as the default because they avoid geographic ambiguity and remain valid even if line routes change due to right-of-way modifications.

Circuit designations must remain stable across line reconfigurations that preserve electrical connectivity. If a line segment is rebuilt with new towers but continues to serve the same terminal substations and protection zones, it should retain its original circuit designation. This stability maintains continuity in protection settings, operational procedures, and compliance documentation. New circuit designations should be assigned only when lines are genuinely new additions to the transmission topology rather than replacements of existing infrastructure.

5.3 Transmission Line Sections and Taps

Transmission lines often include intermediate tap points where power can be delivered to distribution systems or where additional substations connect. These taps create line sections that may require individual identification for protection coordination and fault location purposes. The nomenclature framework handles line sections through subsystem designation: WSTFLD-METRO-345-L1.SEC1

identifies the first section of line L1. The section designation typically reflects the protection zone structure where different line sections operate under different protection relay schemes.

Tapped substations along a transmission line receive identifiers that reference the primary line while indicating the tap configuration: WSTFLD-METRO-345-L1.TAP_INDUST for a tap serving an industrial load. This naming convention captures the topological relationship between the line and the tap and supports automated identification of line configurations relevant to system studies or contingency analysis.

5.4 Transmission Asset Examples

Asset Identifier	Asset Type	Description
WSTFLD-METRO-345-L1	Transmission Line	345kV line 1 between substations
WSTFLD-METRO-345-L2	Transmission Line	345kV line 2 (parallel circuit)
RIVRBND-WSTFLD-345-G1	Generator Lead	345kV generator lead line
CENTRAL-EAST-230-L1.SEC1	Line Section	Section 1 of the 230kV line
WEST-SOUTH-161-L1.TAP_MILL	Line Tap	Tap to Mill substation
NORTH-TERM-500-L3	EHV Line	500kV line 3, long distance
SOLAR-GRID-138-C1	Collection Line	138kV collector from solar farm

6. Substation Asset Nomenclature

Substations contain the most diverse collection of BES assets including power transformers, circuit breakers, disconnect switches, protection relays, potential transformers, current transformers, and control systems. The nomenclature framework must accommodate this diversity while maintaining navigable identifier lengths and supporting both operational usage and automated system queries. The approach employs a hierarchical structure where facility and voltage context establish the top-level namespace, bay assignments provide mid-level organization, and equipment type codes identify specific asset categories.

6.1 Substation Facility Identifiers

Substation facility identifiers follow the same general pattern as other facility types, typically derived from location names or geographic features. Westfield becomes WSTFLD, Metro Central becomes METRO, Riverside becomes RIVERSIDE. The length target of four to eight characters balances brevity with recognition. Substations with names requiring longer abbreviation should prioritize consonants and employ vowel reduction techniques: Shenandoah might become SHNDH, Chattanooga becomes CHTNGA.

Substations serving multiple voltage levels incorporate voltage designation in the system component rather than the facility identifier. This allows the facility identifier to remain consistent across all equipment regardless of voltage association. WSTFLD-345 refers to the 345kV bus at Westfield substation, while WSTFLD-161 refers to the 161kV bus at the same physical facility. This approach supports clear voltage context without proliferating facility identifiers.

6.2 Bay and Position Designations

Substations organize equipment into bays or positions that represent functional groupings of circuit breakers, disconnect switches, and associated protection and control equipment. A transmission line bay contains all equipment associated with terminating a particular transmission line. A transformer bay contains equipment associated with connecting a power transformer. A bus tie bay contains equipment that can connect different bus sections. Bay identifiers should reflect this functional purpose through mnemonic designations: B1, B2, B3 for generic bay numbering; LINE1, LINE2 for transmission line bays; XFMR1, XFMR2 for transformer bays; BUSTIE for bus tie positions.

The bay designation appears as a subsystem component in the hierarchical identifier structure: WSTFLD-345.LINE1.BKR indicates the circuit breaker in line bay 1 at the 345kV bus at Westfield substation. This hierarchical encoding allows automated systems to group all equipment associated with a particular bay through simple string matching operations. All identifiers beginning with WSTFLD-345.LINE1 represent assets within that bay.

6.3 Equipment Type Codes

Specific equipment within substation bays receives type codes that identify the equipment category and support automated queries for equipment inventories and vulnerability assessment. The framework employs standardized codes derived from IEEE and utility industry practice.

Equipment Code	Equipment Type
BKR	Circuit breaker
DISC	Disconnect switch
XFMR	Power transformer
REL	Protection relay (general)
PT	Potential transformer (voltage transformer)
CT	Current transformer
WTG	Wave trap (coupling equipment)
LA	Lightning arrester
CAP	Capacitor bank
REACT	Reactor
BAT	Station battery system
RTU	Remote terminal unit
PANEL	Control panel or cabinet

Equipment codes appear as the final component in substation asset identifiers: WSTFLD-345.LINE1.BKR for the circuit breaker, WSTFLD-345.LINE1.REL-87 for the differential protection relay using ANSI device number 87. This pattern supports both human comprehension and automated parsing where systems can extract equipment type from the identifier component following the bay designation.

6.4 Protection Relay Designations

Protection relays merit special attention in nomenclature due to their critical role in system protection and their complexity as microprocessor-based devices that present cybersecurity exposure. Protection relay identifiers should incorporate ANSI device numbers that indicate relay function: REL-87 for differential protection, REL-21 for distance protection, REL-67 for directional overcurrent protection. This encoding captures protection function information that supports both operational procedures and cybersecurity assessment where different protection functions may employ different relay models with different vulnerability profiles.

Substations employing redundant protection schemes receive relay identifiers that distinguish primary and backup protection: WSTFLD-345.LINE1.REL-87P for primary differential relay, WSTFLD-345.LINE1.REL-87B for backup differential relay. This convention extends to pilot protection schemes requiring relays at both line terminals: WSTFLD-METRO-345-L1.WSTFLD.REL-87 identifies the differential relay at the Westfield terminal of line L1, while WSTFLD-METRO-345-L1.METRO.REL-87 identifies the relay at the Metro terminal.

6.5 Substation Asset Examples

Asset Identifier	Asset Type	Description
WSTFLD-345.LINE1.BKR	Circuit Breaker	Breaker for 345kV line bay 1
WSTFLD-345.LINE1.REL-87P	Diff. Relay	Primary differential protection
WSTFLD-345.LINE1.REL-21	Distance Relay	Distance protection relay
WSTFLD-345.LINE1.DISC-1	Disconnect	Line-side disconnect switch
WSTFLD-345.XFMR1.XFMR	Transformer	345/161kV power transformer
WSTFLD-345.XFMR1.REL-87T	Transformer Relay	Transformer differential relay
WSTFLD-161.BUSTIE.BKR	Bus Tie Breaker	161kV bus tie circuit breaker
WSTFLD-345.LINE1.CT-A	Current Xfmr	Phase A current transformer
WSTFLD-345.LINE1.PT	Potential Xfmr	Voltage sensing transformer
WSTFLD-345.CAP1.CAP	Capacitor Bank	345kV shunt capacitor bank
WSTFLD-161.REACT1.REACT	Reactor	161kV shunt reactor
WSTFLD-SUB.RTU1	RTU	Substation remote terminal unit

7. Control System Asset Nomenclature

Control systems including energy management systems, supervisory control and data acquisition platforms, distributed control systems, and industrial control networks present unique nomenclature challenges. These systems exist as software applications running on computing hardware that may be physically distributed across multiple locations. The nomenclature framework must accommodate both the logical organization of control system applications and the physical topology of computing and network infrastructure. Furthermore, control systems represent the primary attack surface for cyber threats against electric infrastructure, making clear asset identification essential for cybersecurity operations.

7.1 Control Center Facility Identifiers

Control centers receive facility identifiers reflecting their operational scope or geographic service area. A transmission operations center might be identified as TRANSOPS or TOPSCC. A generation control center serving multiple plants might be GENCTL or GENCC. Regional control centers might incorporate geographic designators: WESTCC for western region, CENTRALCC for central region. The facility identifier establishes the namespace for all control system assets within that control center environment.

Backup control centers require distinct facility identifiers to distinguish assets deployed at primary versus backup locations. The pattern FACILITY-BACKUP clearly indicates backup status: TRANSOPS-BACKUP identifies the backup transmission operations center. This distinction matters for both operational procedures and cybersecurity assessment where understanding the physical and logical relationships between primary and backup systems affects incident response planning.

7.2 Control System Application Identifiers

Control system applications receive identifiers that indicate their functional purpose within the control center environment. Standard application designators include EMS for energy management systems, SCADA for supervisory control and data acquisition platforms, DCS for distributed control systems, AGC for automatic generation control, and RTCA for real-time contingency analysis. These mnemonics reflect established industry terminology and enable rapid comprehension by operations and engineering staff.

The application identifier appears as the system component following the facility identifier: TRANSOPS-EMS refers to the energy management system at the transmission operations center. This pattern allows control system identifiers to remain distinct from substation assets while maintaining consistent overall nomenclature structure. Specific application instances receive further designation through subsystem components: TRANSOPS-EMS.SCADA identifies the SCADA component within the EMS application.

7.3 Computing Platform Identifiers

Computing hardware hosting control system applications receives identifiers that capture platform type and application association. Server platforms employ the designation SRV followed by numeric or functional suffixes: TRANSOPS-EMS.SRV-01 for the first EMS server, TRANSOPS-EMS.SCADA-SRV for the dedicated SCADA server. Workstation platforms use WST: TRANSOPS-EMS.WST-OPS1 for operator workstation one. Database servers receive DB designation: TRANSOPS-EMS.DB-SRV for the EMS database server.

Virtualized computing platforms require nomenclature that distinguishes between physical host servers and virtual machine instances. Physical hosts receive identifiers like TRANSOPS-VMHOST-01 to indicate their role as virtualization hosts. Virtual machines include both the host reference and the VM-specific designation: TRANSOPS-VMHOST01.VM-EMS-01 identifies virtual machine EMS-01 running on physical host 01. This hierarchical encoding captures the dependency relationship between virtual and physical infrastructure that matters for both capacity management and incident response.

7.4 Network Infrastructure Identifiers

Network switches, routers, and firewalls supporting control system operations receive identifiers indicating device type and network location. Core network switches at control centers employ designations like TRANSOPS-NET.CORE-SW-01. Access switches connecting end devices use TRANSOPS-NET.ACCESS-SW-01. Firewalls delimiting security zones use TRANSOPS-NET.FW-01 with additional designation indicating protected zones: TRANSOPS-NET.FW-SCADA protects the SCADA network zone.

Wide area network connections between control centers and field sites require identifiers that capture both endpoint locations and circuit function. A SCADA communication circuit from the transmission operations center to Westfield substation might be designated TRANSOPS-WSTFLD-SCADA-CKT1. This pattern identifies both geographic endpoints and the application context, supporting both operational troubleshooting and cybersecurity monitoring where understanding what applications traverse network circuits affects security control selection.

7.5 Control System Asset Examples

Asset Identifier	Asset Type	Description
TRANSOPS-EMS	EMS Application	Energy management system
TRANSOPS-EMS.SCADA-SRV	SCADA Server	SCADA data acquisition server
TRANSOPS-EMS.DB-SRV	Database Server	EMS historian database server
TRANSOPS-EMS.WST-OPS1	Operator Workstation	Operator console 1
TRANSOPS-AGC	AGC Application	Automatic generation control
GENCC-DCS.SRV-01	DCS Server	Distributed control system server
TRANSOPS-VMHOST-01	VM Host	Physical virtualization host
TRANSOPS-VMHOST01.VM-EMS	Virtual Machine	EMS application VM
TRANSOPS-NET.CORE-SW-01	Core Switch	Data center core network switch

TRANSOPS-NET.FW-SCADA	Firewall	SCADA network security firewall
TRANSOPS-WSTFLD-SCADA-CKT1	WAN Circuit	SCADA communication circuit
BACKUP-EMS.STANDBY-SRV	Backup Server	Standby EMS server at backup site

8. Protection System Asset Nomenclature

Protection systems detect abnormal electrical conditions and initiate circuit breaker operations to isolate faulted equipment. Modern protection systems employ microprocessor relays that integrate multiple protection functions, communication capabilities, and sophisticated logic schemes. The nomenclature framework for protection assets extends substation equipment naming conventions while accommodating the functional complexity and cybersecurity significance of modern relay systems.

8.1 Protection Relay Function Designation

Protection relays incorporate ANSI device function numbers that indicate their primary protective purpose. These device numbers have been standard in the electric utility industry for decades and are universally understood by protection engineers. The nomenclature framework requires including ANSI device numbers in protection relay identifiers to capture functional information that supports both operational procedures and automated inventory management for cybersecurity purposes.

ANSI Number	Protection Function
21	Distance protection (impedance-based)
27	Undervoltage protection
32	Directional power protection
50/51	Instantaneous / Time overcurrent
59	Overvoltage protection
67	Directional overcurrent protection
79	Automatic reclosing
81	Frequency protection
87	Differential protection
87T	Transformer differential protection
87L	Line differential protection
87B	Bus differential protection

The ANSI device number appears as a suffix to the relay type designator: WSTFLD-345.LINE1.REL-87L for line differential protection, WSTFLD-345.XFMR1.REL-87T for transformer differential protection. Modern multifunction relays that implement multiple protection functions receive device number designation based on their primary protective function within the protection scheme architecture.

8.2 Redundant Protection Designation

Critical transmission facilities typically employ redundant protection schemes where independent protection systems can detect faults and initiate breaker operations. Redundant protection identifiers distinguish between primary and backup protection relays through P and B suffixes: WSTFLD-345.LINE1.REL-87LP indicates the primary line differential relay, while WSTFLD-345.LINE1.REL-87LB indicates the backup line differential relay. This convention extends to all protection functions where redundancy is deployed.

Pilot protection schemes for transmission lines require protection relays at both line terminals. The nomenclature framework captures this topology through terminal designation in the identifier. For a line between Westfield and Metro substations, the Westfield terminal relay receives identifier WSTFLD-METRO-345-L1.WSTFLD.REL-87L while the Metro terminal relay receives WSTFLD-METRO-345-L1.METRO.REL-87L. This explicit terminal designation supports protection coordination studies and commissioning procedures where understanding relay locations is essential.

8.3 Protection of Communication Assets

Modern protection schemes employ communication systems for pilot protection, breaker failure schemes, and protection coordination. These communication assets require nomenclature that identifies their role within protection architectures. Fiber-optic communication equipment supporting differential protection is assigned identifiers such as WSTFLD-345.LINE1.FIBER-MUX indicates the fiber multiplexer for line 1. Direct transfer trip communication channels are designated WSTFLD-METRO-345-L1.DTT-CH1 for direct transfer trip channel 1.

8.4 Protection System Asset Examples

Asset Identifier	Asset Type	Description
WSTFLD-345.LINE1.REL-87LP	Diff. Relay	Primary line differential relay
WSTFLD-345.LINE1.REL-87LB	Diff. Relay	Backup line differential relay
WSTFLD-345.LINE1.REL-21	Distance Relay	Distance protection relay
WSTFLD-345.LINE1.REL-67	Dir. OC Relay	Directional overcurrent relay
WSTFLD-345.XFMR1.REL-87TP	Xfmr Diff. Relay	Primary transformer diff. relay
WSTFLD-345.BUS.REL-87B	Bus Diff. Relay	Bus differential relay
WSTFLD-METRO-345-L1.WSTFLD.REL-87L	Line Terminal Relay	Westfield terminal relay
WSTFLD-METRO-345-L1.METRO.REL-87L	Line Terminal Relay	Metro terminal relay

WSTFLD-345.LINE1.FIBER-MUX	Fiber Equipment	Protection fiber multiplexer
WSTFLD-METRO-345-L1.DTT-CH1	Transfer Trip	Direct transfer trip channel
WSTFLD-345.LINE1.REL-79	Reclose Relay	Automatic reclosing relay
WSTFLD-345.LINE1.GPS-CLK	GPS Clock	GPS time synchronization

9. Communication Asset Nomenclature

Communication systems supporting BES operations include microwave systems, fiber optic networks, power line carrier systems, and radio networks. These systems carry SCADA data, protection signaling, voice communications, and increasingly, IP-based applications including video surveillance and remote access. Communication asset nomenclature must accommodate the diversity of communication technologies while supporting operational troubleshooting and cybersecurity monitoring that requires understanding which applications traverse which communication paths.

9.1 Communication System Identifiers

Communication systems receive facility-level identifiers that reflect their technology and service area. A microwave system serving the transmission operations center might be designated TRANSOPS-MW for microwave. Fiber optic systems are designated with FIBER: TRANSOPS-FIBER indicates the fiber optic network serving transmission operations. Radio systems use RADIO designation: TRANSOPS-RADIO-VHF for VHF radio system serving transmission operations.

9.2 Communication Path Identifiers

Individual communication paths between facilities require identifiers that capture both endpoint locations and the communication technology employed. The basic pattern follows ORIGIN-DESTINATION-TECHNOLOGY-CIRCUIT. A microwave path from transmission operations to Westfield substation would be TRANSOPS-WSTFLD-MW-PATH1. A fiber path follows the same pattern: WSTFLD-METRO-FIBER-PATH1 for fiber connecting Westfield to Metro substations.

9.3 Communication Equipment Identifiers

Communication equipment at facilities receives identifiers indicating equipment type and communication path association. Microwave radios are designated MW-RADIO: TRANSOPS-MW.RADIO-01 for microwave radio 01 at the transmission operations center. Fiber multiplexers use FIBER-MUX: WSTFLD-345.FIBER-MUX-01 for fiber multiplexer at Westfield substation. Channel banks converting between analog and digital signals use CHBANK: TRANSOPS-CHBANK-01. These identifiers support both operational procedures and inventory management for communication asset tracking.

9.4 Communication Asset Examples

Asset Identifier	Asset Type	Description
TRANSOPS-MW	MW System	Transmission operations microwave system
TRANSOPS-FIBER	Fiber System	Transmission operations fiber network
TRANSOPS-WSTFLD-MW-PATH1	MW Path	Microwave path to substation
WSTFLD-METRO-FIBER-PATH1	Fiber Path	Inter-substation fiber path

TRANSOPS-MW.RADIO-01	MW Radio	Microwave radio terminal
WSTFLD-345.FIBER-MUX-01	Fiber Mux	Fiber multiplexer equipment
TRANSOPS-RADIO-VHF	Radio System	VHF radio communication system
WSTFLD-SUB.RADIO-BASE	Radio Base	Substation radio base station
TRANSOPS-PLC-CH1	PLC Channel	Power line carrier channel
WSTFLD-345.LINE1.PLC-XMTR	PLC Transmitter	Power line carrier transmitter
TRANSOPS-SAT.TERMINAL	Satellite Term	Satellite communication terminal
TRANSOPS-CHBANK-01	Channel Bank	Voice channel bank equipment

10. Delimiter Conventions and Parsing Rules

The effectiveness of a nomenclature framework depends not only on the information encoded in asset identifiers but also on the systematic structure that enables both human comprehension and automated parsing. This section codifies the delimiter conventions and parsing rules that underpin the entire nomenclature architecture. Understanding these conventions is essential for both operational personnel who must interpret asset identifiers and for developers implementing automated systems that parse identifier components.

10.1 Primary Delimiter: Hyphen

The hyphen character serves as the primary delimiter separating major identifier components that represent different hierarchy levels or orthogonal classification dimensions. In the identifier WSTFLD-345-LINE1-BKR, hyphens separate the facility identifier (WSTFLD), voltage class (345), functional system (LINE1), and equipment type (BKR). Each hyphen-delimited component represents a distinct classification axis essential for asset identification. Automated systems can extract these components through simple string splitting operations on the hyphen character.

Hyphen usage follows strict conventions that maintain parsing consistency. Hyphens appear only as delimiters between major components; they never appear within component values except in the specific case of compound voltage designations like 345-230 indicating transformation between voltage levels. This consistency eliminates ambiguity in parsing operations where systems can rely on hyphen characters as definitive component boundaries.

10.2 Secondary Delimiter: Period

The period character serves as a secondary delimiter for sub-components within a hierarchical level. In the identifier WSTFLD-345.LINE1.BKR, the period separates voltage-level context (345) from bay designation (LINE1) and equipment type (BKR). This hierarchical encoding allows a single identifier to capture both the major classification dimension (the 345kV voltage level) and the nested containment relationships within that dimension (bay LINE1 containing breaker BKR).

Period delimiters enable representation of hierarchies of arbitrary depth while maintaining parseability. An identifier like WINDS-WIND-NORTH.WT15.CTRL.PLC captures four hierarchical levels: zone (NORTH), turbine (WT15), system (CTRL), component (PLC). Automated systems extract this hierarchy through recursive period-based parsing, enabling queries like 'all assets in zone NORTH' or 'all control PLCs across all turbines.'

10.3 Tertiary Delimiter: Underscore

The underscore character joins multi-word descriptors that function as single logical units. In TRANSOPS-EMS.SCADA_SRV, the underscore indicates that SCADA_SRV is a single functional component (the SCADA server) rather than separate SCADA and SRV components. This convention prevents multi-word component names from being misinterpreted as separate hierarchy levels. The underscore signals to both human readers and automated parsers that words connected by underscores should be treated as atomic units during component extraction.

Underscore usage follows the convention that spaces in component names should be replaced with underscores rather than removed entirely. A facility named 'North Valley' becomes NORTH_VALLEY rather than NORTHVALLEY, preserving word boundaries that aid human comprehension. Automated systems treat underscores as internal to component values, not as delimiters for parsing operations. The distinction matters for parsing rules where NORTH_VALLEY-345 splits on the hyphen into components 'NORTH_VALLEY' and '345', with the underscore preserved within the first component.

10.4 Character Set Constraints

Asset identifiers must conform to strict character set limitations to ensure compatibility with diverse information systems including databases, SCADA systems, protection relay configuration tools, and document management systems. The permitted character set includes uppercase letters A through Z, digits 0 through 9, hyphens, periods, and underscores. Lowercase letters are prohibited to eliminate case sensitivity issues in systems that may or may not preserve case. Special characters beyond the three defined delimiters are prohibited to prevent parsing ambiguity and database compatibility problems.

Spaces are explicitly prohibited in asset identifiers. Where component names naturally include spaces, those spaces must be replaced with underscores following the convention established above. This prohibition on spaces eliminates an entire class of parsing problems arising from inconsistent space handling across different systems. It also prevents issues with command-line tools, URL encoding, and database query syntax where spaces require special escaping or quoting.

10.5 Parsing Algorithm Specification

Automated systems should parse BES asset identifiers using the following algorithmic approach. First, split the identifier on hyphen characters to extract major components. Each hyphen-delimited component represents either a distinct classification attribute or a hierarchical context. Second, for components that contain periods, split on period characters to extract hierarchical sub-components. The resulting hierarchy depth varies by asset type but follows consistent patterns within each category. Third, preserve underscores within components as internal characters that form part of component values rather than structural delimiters.

Consider the identifier WSTFLD-345.LINE1.REL-87LP. First-level parsing on hyphens produces three major components: 'WSTFLD', '345.LINE1.REL', and '87LP'. Second-level parsing of the middle component on periods produces the hierarchical structure: '345' (voltage level), 'LINE1' (bay), 'REL' (equipment type). The result is a parsed structure with facility WSTFLD, voltage 345, bay LINE1, equipment type REL, and function 87LP. This structure supports queries at any hierarchy level from facility-wide inventories down to specific equipment instances.

11. Classification Metadata Encoding

Asset identifiers serve dual purposes as both operational references and metadata containers. The nomenclature framework embeds classification information directly in identifier structure, enabling automated systems to extract asset attributes without database lookups. This section defines the specific metadata elements encoded in identifiers and their operational significance for compliance tracking, cybersecurity assessment, and operational coordination.

11.1 Asset Type Classification

The equipment type component of asset identifiers explicitly indicates the category of physical or logical asset represented. This classification supports automated queries for asset inventories required by NERC compliance audits and cybersecurity assessments. When a vulnerability advisory indicates that a particular relay model is affected, systems can automatically identify all protection relays by searching for identifiers containing 'REL' as an equipment type component. The classification granularity balances the specificity needed for operational purposes against the practical limits of mnemonic brevity.

11.2 Voltage Class Metadata

Voltage class designation in transmission and substation asset identifiers enables automated classification of assets by their operational characteristics and regulatory applicability. NERC standards apply different requirements to facilities operating at different voltage levels. The CIP standards define Bulk Electric System assets partly through voltage thresholds. Embedding voltage class directly in identifiers allows compliance tracking systems to automatically determine which standards apply to which assets through simple identifier parsing rather than requiring manual classification or database attribute management.

Voltage class metadata also supports operational procedures where voltage determines fundamental operating constraints and protection requirements. Automated work management systems can use voltage information extracted from asset identifiers to apply appropriate safety procedures, clearance requirements, and isolation protocols. Similarly, contingency analysis systems can classify assets by voltage to assess system impacts of potential outages at different system levels.

11.3 Functional System Context

The system component of asset identifiers captures functional context that indicates the operational purpose and technical characteristics of asset groups. For control systems, functional designators like EMS, SCADA, and AGC indicate the application domain, enabling automated correlation of cybersecurity threats to affected systems. When a vulnerability emerges in SCADA software, systems can identify all SCADA-related assets through identifier patterns. For generation assets, technology codes like NGCC, COAL, and SOLAR_PV support automated reporting on generation fleet composition and emissions tracking.

11.4 Geographic and Topological Context

Facility identifiers and transmission line terminal designations embed geographic context that supports operational coordination and incident response. When a severe weather event threatens a geographic

region, operations centers can automatically identify potentially affected assets via facility-identifier matching. For transmission lines, encoding both terminal substations enables automated identification of all lines connecting to a particular substation, supporting operational procedures for substation outages that require reconfiguration of multiple transmission circuits.

11.5 Ownership and Jurisdictional Metadata

Entity identifiers provide ownership attribution, which is essential for compliance responsibilities and coordination across organizational boundaries. In interconnected transmission systems where a single line may have segments owned by different entities, the nomenclature framework supports compound identifiers that capture ownership transitions. For jointly-owned facilities, naming conventions should establish clear ownership attribution through primary entity designation while documenting ownership complexity in facility metadata systems referenced by the identifier.

12. Integration with Compliance and Cybersecurity Systems

The nomenclature framework directly supports integration with automated compliance evidence collection and cybersecurity operations. This section defines how standardized asset identifiers enable system capabilities ranging from NERC audit evidence automation to real-time threat intelligence correlation. These integration patterns demonstrate the operational value proposition for nomenclature standardization investments.

12.1 ERT Companion Evidence Collection

The ERT Companion system leverages standardized asset nomenclature to automate the collection of evidence required for NERC compliance audits. When auditors issue evidence requests requiring documentation of Electronic Security Perimeters, the system automatically correlates ESP boundary definitions to affected assets through identifier pattern matching. Assets whose identifiers include control center facility codes and control system application designators are automatically associated with ESP documentation. This automation eliminates the manual effort historically required to assemble asset inventories for audit responses.

Asset identifier standardization enables the ERT Companion to validate completeness of evidence submissions through automated cross-referencing. If an entity reports operating fifteen substations but evidence submissions reference assets at only twelve substation facility codes, the system automatically flags the discrepancy for investigation. This validation capability reduces compliance risk arising from inadvertent omissions in evidence documentation.

12.2 Analyst's WorkBench Threat Correlation

The Analyst's WorkBench cyber threat intelligence platform uses asset nomenclature metadata to automate the correlation of vulnerability advisories with affected infrastructure. When a vendor publishes a security advisory affecting a particular protection relay model, the AWB automatically identifies all protection relays in the entity's infrastructure through queries for identifiers containing 'REL' equipment type codes. Cross-referencing relay identifiers with asset management system records that document relay models enables automated generation of affected asset lists for incident response teams.

The nomenclature framework supports AWB's strategic compression-mitigation capabilities by enabling rapid translation from threat-intelligence abstractions into concrete infrastructure-impact assessments. When threat intelligence reports emerge describing exploitation of SCADA systems, AWB automatically identifies SCADA infrastructure through facility and system identifier patterns. This automation reduces the time required to assess organizational exposure from hours or days of manual analysis to seconds of automated correlation.

12.3 Configuration Management and Change Control

Standardized asset nomenclature enables automated configuration management where system changes must be coordinated across multiple documentation systems. When a protection relay at a substation is replaced, change management systems use the asset identifier as a correlation key to automatically identify all documentation requiring updates, including single-line diagrams, protection coordination

studies, communication circuit records, and asset management system records. This automation reduces configuration drift where documentation systems fall out of synchronization due to incomplete change propagation.

12.4 Vendor Coordination and Supply Chain Management

Asset identifiers serve as stable references for vendor coordination across the equipment lifecycle from procurement through decommissioning. Purchase orders, warranty documentation, and maintenance contracts reference assets through standardized identifiers that remain valid across organizational changes and system migrations. When vendors issue field notices or service bulletins, entities can automatically identify affected assets by correlating vendor equipment records with asset identifiers. This capability becomes increasingly important as supply chain cybersecurity requirements mandate tracking of equipment provenance and patch status across entire infrastructure portfolios.

13. Implementation of Guidelines and Transition Strategies

Implementing nomenclature standardization across an operating electric utility requires careful planning to balance operational continuity with the benefits of standardization. This section provides pragmatic guidance for phased implementation approaches that minimize operational disruption while establishing consistent naming practices for both new construction and existing infrastructure, where transition is feasible.

13.1 New Asset Naming Requirements

All newly commissioned BES assets must conform to the nomenclature framework defined in this specification. This requirement applies to new generation facilities, transmission lines, substations, and control system implementations. The compliance mandate for new assets establishes a foundation for standardization without requiring disruptive renaming of existing operational infrastructure. New asset compliance should be verified through design review processes before commissioning to prevent non-conforming identifiers from entering operational use.

Equipment replacements at existing facilities present opportunities for selective nomenclature updates where feasible. When a protection relay is replaced at a substation, the new relay should receive an identifier conforming to current standards even if surrounding equipment retains legacy naming. This incremental approach gradually improves nomenclature consistency without requiring facility-wide renaming campaigns. Documentation systems must support maintenance of both legacy and standardized identifiers during transition periods through alias mechanisms that map legacy names to current standards.

13.2 Legacy Asset Transition Approaches

Existing assets with legacy naming conventions should be evaluated for renaming feasibility based on the operational impact of identifier changes. Critical consideration factors include: protection relay settings that embed asset names, SCADA system configurations with hardcoded references, and operator training investments in legacy naming schemes. Assets where renaming requires extensive system reconfiguration and operator retraining should retain legacy identifiers while implementing alias mappings in documentation systems.

Major facility modifications or system upgrades create natural opportunities for legacy nomenclature transitions. When a substation undergoes comprehensive modernization replacing multiple equipment types, the project scope can include identifier standardization across all affected assets. The concentrated change management effort required for major modifications already necessitates documentation updates and operator retraining, making nomenclature transition feasible with incremental additional effort.

13.3 Documentation System Requirements

Information systems supporting BES operations must accommodate standardized asset identifiers through appropriate database schema and user interface design. Asset management systems should implement identifier fields conforming to the character set and structure requirements defined in this specification. SCADA systems must support identifier patterns including hierarchical period-delimited

structures. Protection relay configuration tools should validate identifiers against nomenclature rules during commissioning processes.

Documentation systems must support identifier alias mechanisms during transition periods where legacy and standardized naming coexist. Database implementations should maintain both legacy and current identifiers with automatic translation capabilities enabling cross-referencing. User interfaces should display current standardized identifiers while supporting search and filtering operations on legacy aliases. This dual-identifier support minimizes operational disruption during transition while establishing standardized naming as the authoritative reference.

13.4 Training and Change Management

Personnel training on nomenclature standards should emphasize the operational benefits of standardization rather than treating naming conventions as arbitrary administrative requirements. Operators, protection engineers, and maintenance staff should understand how standardized identifiers support operational procedures, troubleshooting, and compliance documentation. Training should include practical exercises in identifier interpretation and construction for common asset types to build fluency with nomenclature patterns.

Change management communications should acknowledge the operational burden of nomenclature transitions while articulating concrete benefits including automated compliance evidence collection, improved cybersecurity threat correlation, and reduced ambiguity in operational communications. Executive sponsorship for nomenclature standardization should emphasize alignment with broader digital transformation initiatives including control system modernization and cybersecurity program enhancement.

14. Governance and Standards Maintenance

Nomenclature standards require ongoing governance to address emerging technologies, operational practice evolution, and organizational learning from implementation experiences. This section establishes governance structures and change management processes that maintain standard relevance while preserving the stability essential for operational systems that depend on consistent asset identification.

14.1 Standards Authority and Change Control

A designated standards authority within each registered entity should maintain responsibility for nomenclature framework governance including interpretation of ambiguous cases, approval of extensions for novel asset types, and periodic standard revisions. The standards authority should include representation from operations, protection engineering, cybersecurity, and compliance functions to ensure decisions reflect diverse operational requirements. Changes to core nomenclature rules require approval through formal change control processes that assess operational impact and implementation feasibility.

14.2 Extension Mechanisms for Novel Technologies

The framework provides extension mechanisms for emerging technologies not explicitly addressed in current specifications. When novel asset types emerge requiring naming conventions, entities should propose extensions following established patterns for similar asset categories. Battery energy storage systems follow generation asset patterns with technology code BATTERY. Hydrogen fuel cells might employ technology code H2-FC following the same structure. Extension proposals should document operational requirements, demonstrate consistency with framework principles, and identify integration points with compliance and cybersecurity systems.

14.3 Periodic Review and Revision Cycles

The nomenclature specification should undergo formal review on a biennial cycle to incorporate lessons learned from implementation experiences and address technology evolution. Review processes should solicit input from operational staff, protection engineers, and cybersecurity personnel to identify ambiguities, operational friction points, and opportunities for clarification. Revisions should prioritize backward compatibility where feasible, with breaking changes reserved for situations where operational benefits clearly justify transition costs.

14.4 Industry Coordination and Harmonization

While this specification addresses requirements for individual registered entities, industry benefit would accrue from nomenclature harmonization across organizational boundaries. Regional transmission organizations and reliability coordinators should consider adoption of consistent nomenclature frameworks that facilitate coordination across multiple entities. Industry working groups might develop recommended practices for multi-entity facilities and jointly-owned infrastructure where consistent naming supports operational coordination and regulatory compliance across ownership boundaries.

15. Conclusion

The BES asset nomenclature framework presented in this specification establishes systematic identification conventions that serve operational, compliance, and cybersecurity objectives while accommodating the diversity of asset types and operational contexts present across electric utility infrastructure. The framework balances standardization benefits against operational realities by tiering requirements: it mandates consistency in core identification elements while providing flexibility for entity-specific practices in non-critical components.

Successful implementation requires commitment to applying nomenclature standards to new asset commissioning while pragmatically managing legacy naming through alias mechanisms and selective transition opportunities. The operational benefits of standardization become apparent through integration with automated systems, including NERC compliance evidence collection, cybersecurity threat intelligence correlation, and operational coordination across organizational boundaries. These capabilities justify the investment required to establish consistent naming practices across large, complex infrastructure portfolios.

The nomenclature framework enables the digital transformation of electric utility operations by providing a foundational identification layer on which automated compliance, cybersecurity, and operational systems depend. As utilities continue to integrate information technology and operational technology, systematic asset identification becomes increasingly critical for maintaining situational awareness across a more complex, interconnected infrastructure. The investment in nomenclature standardization is essential to utility capabilities, ranging from automated evidence collection to AI-enabled operational intelligence, that will characterize the industry's evolution over the coming decades.

Appendix A: Quick Reference Tables

A.1 Generation Technology Codes

Code	Technology
COAL	Coal-fired steam generation
NGCC	Natural gas combined-cycle
CT	Combustion turbine simple-cycle
NUCLEAR	Nuclear steam generation
HYDRO	Conventional hydroelectric
PUMPED	Pumped storage hydroelectric
WIND	Wind turbine generation
SOLAR-PV	Photovoltaic solar
SOLAR-T	Solar thermal
BATTERY	Battery energy storage
DIESEL	Diesel reciprocating engine
GEO	Geothermal generation

A.2 Substation Equipment Type Codes

Code	Equipment Type
BKR	Circuit breaker
DISC	Disconnect switch
XFMR	Power transformer
REL	Protection relay
PT	Potential transformer
CT	Current transformer
RTU	Remote terminal unit
CAP	Capacitor bank
REACT	Shunt reactor
LA	Lightning arrester

BAT	Station battery
PANEL	Control panel

A.3 Protection Relay ANSI Device Numbers

ANSI Number	Function
21	Distance protection
27	Undervoltage
50/51	Overcurrent
59	Overvoltage
67	Directional overcurrent
79	Automatic reclosing
81	Frequency
87	Differential protection
87L	Line differential
87T	Transformer differential
87B	Bus differential

A.4 Control System Application Codes

Code	Application Type
EMS	Energy management system
SCADA	Supervisory control and data acquisition
DCS	Distributed control system
AGC	Automatic generation control
RTCA	Real-time contingency analysis
PLC	Programmable logic controller
HMI	Human-machine interface

A.5 Standard Voltage Classes

Voltage (kV)	Typical Application
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765	Extra-high voltage transmission
500	Extra-high voltage transmission
345	High voltage transmission
230	High voltage transmission
161	High voltage transmission
138	High voltage transmission
115	High voltage transmission
69	Sub-transmission / distribution

A.6 Delimiter Usage Summary

Delimiter	Usage
Hyphen (-)	Separates major identifier components
Period (.)	Separates hierarchical sub-components
Underscore (_)	Joins multi-word component names
None (prohibited)	Spaces and special characters

Appendix B: Implementation Checklist

This checklist provides a structured approach for entities implementing the BES asset nomenclature framework. Items are organized by implementation phase to support systematic deployment while maintaining operational continuity.

Phase 1: Planning and Preparation

Establish standards governance authority with representation from operations, protection, cybersecurity, and compliance. Review existing asset naming conventions and document extent of standardization required. Identify critical systems with hardcoded asset references requiring migration planning. Assess database schema requirements for identifier field modifications. Develop training materials explaining nomenclature principles and operational benefits. Create implementation timeline with milestones for new asset requirements and legacy transition phases.

Phase 2: Database and System Preparation

Modify asset management system schema to support standardized identifier formats including hierarchical period-delimited structures. Implement alias table mechanisms for mapping legacy identifiers to standardized equivalents. Update SCADA system configuration tools to validate identifier patterns during commissioning. Configure protection relay setting software to enforce identifier conventions. Deploy documentation management systems supporting dual-identifier display during transition. Establish automated validation tools that check identifier conformance during asset registration.

Phase 3: Initial Deployment

Issue directive requiring nomenclature compliance for all newly commissioned assets. Integrate identifier validation into project design review processes. Update standard drawings and templates to reflect nomenclature conventions. Conduct training sessions for engineering staff responsible for asset commissioning. Commission pilot installations demonstrating nomenclature framework implementation. Document lessons learned and refine implementation procedures based on initial experience.

Phase 4: Legacy Transition

Identify major facility modifications providing opportunities for selective legacy nomenclature updates. Develop facility-specific transition plans assessing operational impact of identifier changes. Implement alias mappings for legacy identifiers in all documentation systems. Update operator training materials reflecting identifier transitions. Coordinate protection relay setting modifications with nomenclature updates where feasible. Monitor operational impacts and adjust transition pace based on organizational capacity.

Phase 5: Integration Enablement

Configure ERT Companion evidence collection to leverage standardized asset identifiers for automated compliance documentation. Deploy Analyst's WorkBench threat correlation capabilities using

nomenclature metadata for affected asset identification. Integrate asset identifiers with vendor management systems for supply chain tracking. Implement automated configuration management leveraging identifiers as correlation keys across documentation systems. Enable operational analytics using identifier metadata for asset classification and trend analysis.